Conway's 99-graph: the involution case

Exact reduction, an exact $\mathbb{F}_2$-rank identity, an audit of the order-seven census, and first computations

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operated by D. Djematene; every claim is either cited to a primary source or proved below

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Abstract

Let Γ be a putative strongly regular graph $\mathrm{srg}(99, 14, 1, 2)$ (Conway's 99-graph). We (i) record the exact state of the automorphism programme from primary sources, which leaves $|\mathrm{Aut}(\Gamma)| \in \{1, 2, 3\}$ and shows that the one-fixed-point $\mathbb{Z}_7$ case, described as open in recent work, was in fact closed at the orbit-matrix level in 2009/2011; (ii) show that the involution case is completely rigid and reduce it to an exact finite problem on 42 orbits, with a perfect matching structure; (iii) derive an exact $\mathbb{F}_2$-rank equation for the graph of single adjacencies (a consequence of the quotient equations, useful as a filter, not an independent obstruction); (iv) audit the order-seven induced-subgraph census of Reimbayev (arXiv:2608.19410) and locate four typographical errors, after which the census is exactly equivalent to a system already verified in the public repository `conway-99-research` and yields no new constraint; (v) report the first exact computations on the involution case (validated SAT/CP-SAT models; all runs so far inconclusive). Nothing here resolves the existence problem.

1 Status of the symmetry programme (primary sources)

- Makhnev–Minakova [1] (Diskr. Mat. 16:1, 2004; Theorem 1 verbatim): if $g \in \mathrm{Aut}(\Gamma)$ has prime order, then $\mathrm{Fix}(g)$ is a single vertex and $|g| \in \{2, 7\}$, or empty and $|g| \in \{3, 11\}$, or a triangle and $|g| = 3$; moreover $|\mathrm{Aut}(\Gamma)|$ divides $2 \cdot 3^3 \cdot 7 \cdot 11$.
- Behbahani [2] (Thm. 4.14, Table 21) = Behbahani–Lam [3]: by exhaustive orbit-matrix enumeration, no orbit matrix exists for $p \in \{5, 7, 11\}$ (all admissible fixed-point counts: $p = 7$ with $1, 8, 15, 22$ fixed points; $p = 11$ with $0, 11, 22$), nor for $p = 3$ with fixed points. Hence the prime divisors of $|\mathrm{Aut}(\Gamma)|$ are 2 and 3, and an element of order 3 is fixed-point-free. **The case $p = 2$ was not computed** (the table starts at $p = 3$).
- Crnković–Maksimović [4]: no automorphism group of order 6 or 9. Cesarz–Woldar [5] (computer-free): $2 \mid |G| \Rightarrow |G| \mid 6$; $7 \mid |G| \Rightarrow G \cong \mathbb{Z}_7$.
- Consequently $|\mathrm{Aut}(\Gamma)| \in \{1, 2, 3\}$. Thakkar–Severini [6] write that the single-fixed-point $\mathbb{Z}_7$ case "remains open"; by [2, 3] it is closed. The fixed-point-free order-3 branch is currently pursued by several public projects; the involution branch has, to our knowledge, never been the object of an exact computation.
- Ishida [7] recovers $a_1(t) = 14$ for an involution t from a trace–rank identity; no exclusion.

2 Rigidity of the involution

Let $t \in \mathrm{Aut}(\Gamma)$ be an involution; by [1], $\mathrm{Fix}(t) = \{x\}$. Root Γ at x: $N(x)$ induces a perfect matching $7K_2$ with blocks $B_i = \{(i, 0), (i, 1)\}$, $i = 1, \ldots, 7$. Every vertex at distance 2 has exactly two neighbours in $N(x)$ (as $\mu = 2$), which lie in different blocks (as $\lambda = 1$), and the 84 outer vertices are in bijection with the 84 unordered pairs $\{(i, s), (j, e)\}$, $i < j$ (the classical rooted

reduction, written out e.g. in [5], [10] and [6]). The only unknown is the 12-regular outer graph H.

Lemma 1 is essentially Makhnev's lemma $|\operatorname{Fix}(t)| = 1 + |\operatorname{Fix}(t) \cap [a]| + 2l$ [9] (with $\operatorname{Fix}(t) = \{x\}$, $l = 0$), and $a_1(t) = 14$ is in [1]; what appears to be new here is Proposition 3 and the exact model built on it.

Lemma 1. *t acts on the inner vertices as the flip $\sigma : (i, s) \mapsto (i, 1 - s)$ in every block, and on labels by $\sigma\{a, b\} = \{a', b'\}$.*

Proof. t fixes x, hence preserves $N(x)$ and its matching, and fixes no inner vertex. If t swapped two inner vertices a, b from different blocks, it would fix the label $\{a, b\}$, an outer vertex, contradicting $\operatorname{Fix}(t) = \{x\}$. So every block is flipped. □

Lemma 2. *No vertex other than the 14 inner ones is adjacent to its image. In particular $a_1(t) = 14$ and $\chi_3(t) = \chi_{-4}(t) = 0$.*

Proof. If $u \sim tu$, the edge $\{u, tu\}$ is t-invariant, so its unique common neighbour ($\lambda = 1$) is t-fixed, hence equals x, hence $u \in N(x)$. With $a_0 = 1$, $a_1 = 14$, $a_2 = 84$ and $E_3 = \frac{4}{7}I + \frac{1}{7}A - \frac{2}{77}J$: $\chi_3(t) = \frac{6}{11} + 14 \cdot \frac{9}{77} - 84 \cdot \frac{2}{77} = 0$, and $\chi_{-4}(t) = 1 - 1 - \chi_3(t) = 0$. □

Proposition 3. *For every outer u, $N(u) \cap N(\sigma u)$ is a σ-orbit $\{v, \sigma v\} =: D(u)$, and D induces a perfect matching on the 42 σ-orbits of outer vertices. Each outer vertex is adjacent to both members of exactly one orbit.*

Proof. $u \not\sim \sigma u$ by Lemma 2, so $|N(u) \cap N(\sigma u)| = \mu = 2$; the set is σ-invariant and contains no fixed point, hence is an orbit. If $D(u) = \{v, \sigma v\}$ then $u, \sigma u \in N(v) \cap N(\sigma v)$, so $D(v) = \{u, \sigma u\}$. □

Orbit-level exact model. For orbits $U \neq V$ with representatives u, v put $(a_{UV}, b_{UV}) = (A[u, v], A[u, \sigma v]) \in \{0, 1\}^2$; the pair is symmetric in U, V. Given the forced part, $A^2 + A - 12I = 2J$ is equivalent to: (R) $\sum_V (a_{UV} + b_{UV}) = 12$; (D) exactly one V with $a_{UV} = b_{UV} = 1$; (I) for every inner vertex c, the number of H-neighbours of u whose label contains c is 1 if $c \in u$ or $c' \in u$ and 2 otherwise; (P) for $U \neq V$: $\sum_W (a_{UW}a_{VW} + b_{UW}b_{VW}) + a_{UV} = 2 - |u \cap v|$ and $\sum_W (a_{UW}b_{VW} + b_{UW}a_{VW}) + b_{UV} = 2 - |u \cap \sigma v|$.

Quotients. Let Q (50×50) be the ordinary orbit matrix and S (49×49) the matrix of A on the (-1)-eigenspace of t in the basis $e_u - e_{\sigma u}$. Then $Q^2 + Q - 12I = 2N$ ($N_{UV} = |V|$), $Q_{\text{out}} = \Gamma_1 + 2D$ with Γ_1 10-regular, and $S^2 + S = 12I$ exactly, $\operatorname{Spec}(S) = \{3^{27}, (-4)^{22}\}$. Writing $S = \begin{pmatrix} -I_7 & C^{\mathsf{T}} \\ C & S_{\text{out}} \end{pmatrix}$ with $C^{\mathsf{T}}C = 12I_7$: $C^{\mathsf{T}}S_{\text{out}} = 0$, $S_{\text{out}}^2 + S_{\text{out}} = 12I - CC^{\mathsf{T}}$, $S_{\text{out}} \in \{0, \pm 1\}^{42 \times 42}$ with ten nonzeros per row, $\operatorname{Spec}(S_{\text{out}}) = \{0^7, 3^{20}, (-4)^{15}\}$, and $(S_{\text{out}} + 4I - \frac{1}{3}CC^{\mathsf{T}})/7$ is a projector of rank 20.

3 An exact $\mathbb{F}_2$-rank identity

Theorem 4. *Let Γ admit an involution, and let $\bar{\Gamma} \in \mathbb{F}_2^{42 \times 42}$ be the adjacency matrix of Γ_1 (the graph of single adjacencies on the 42 outer orbits) and $N_b \in \mathbb{F}_2^{42 \times 7}$ the orbit/block incidence. Then*

$$\bar{\Gamma}^2 + \bar{\Gamma} = N_b N_b^{\mathsf{T}} \quad \text{and} \quad \operatorname{rank}_{\mathbb{F}_2}\big(\bar{\Gamma} + N_b N_b^{\mathsf{T}}\big) = \operatorname{rank}_{\mathbb{F}_2}(\bar{\Gamma}^2) = 20.$$

Equivalently $K := \bar{\Gamma}^2$ is a symmetric idempotent of rank exactly 20 with $KN_b = 0$, and $\bar{\Gamma} = K + N_b N_b^{\mathsf{T}}$.

Proof. The signed quotient S satisfies $S^2 + S = 12I$, so its eigenvalues are 3 and -4, with multiplicities f, g satisfying $f + g = 49$ and $3f - 4g = \operatorname{tr}(S) = -7$ (inner block $-I_7$, outer diagonal zero by Lemma 2); hence $f = 27$, $g = 22$, and the characteristic polynomial of S

is $(x-3)^{27}(x+4)^{22}$. Reducing the integer matrix S modulo 2 gives characteristic polynomial $(x+1)^{27}x^{22}$, and $S^2+S=12I\equiv 0$ shows that $\bar{S}:=S \bmod 2$ is diagonalizable over $\mathbb{F}_2$; therefore $\operatorname{rank}_{\mathbb{F}_2}(\bar{S})=27$. Since $\bar{S}=\begin{pmatrix} I_7 & N_b^{\mathsf{T}} \\ N_b & \bar{\Gamma} \end{pmatrix}$, a Schur complement gives $\operatorname{rank}_{\mathbb{F}_2}(\bar{\Gamma}+N_bN_b^{\mathsf{T}})=20$. The identity $\bar{\Gamma}^2+\bar{\Gamma}=N_bN_b^{\mathsf{T}}$ is the reduction mod 2 of $Q_{\text{out}}^2+Q_{\text{out}}=12I+4J-\text{Sh}$ (where Sh_{UV} is the number of shared blocks). Finally $N_b^{\mathsf{T}}N_b\equiv 0$, so $(N_bN_b^{\mathsf{T}})^2=0$ and $\bar{\Gamma}N_b=0$ (every vertex has an even number of single neighbours on each block), whence $K^2=K$ and $KN_b=0$. □

Remark 5. A first version of this proof went through the 2-adic eigenlattices ($L_3\cong\mathbb{Z}_2[C_2]^{27}$, $L_{-4}\cong\mathbb{Z}_2[C_2]^{22}$ free, via Broué's theorem on p-permutation lattices); the elementary derivation above, pointed out by an independent check, needs only the spectrum of S and a Schur complement. The mod-2 pairwise identities alone only yield $K^2=K$ and $\operatorname{rank}K$ even. However the value 20 follows from the integer identity $S^2+S=12I$ and $\operatorname{tr}S=-7$, which are consequences of the exact model below; it is therefore *not* an independent obstruction, but a global invariant usable as a (late) filter in a generator. The analogue at $k=22$ ($\operatorname{rank}_{\mathbb{F}_2}(\bar{\Gamma}+N_bN_b^{\mathsf{T}})=44$) is verified on the Berlekamp–van Lint–Seidel graph (`code/z2/control_bvls.py`). Mod 7 one finds that $S_{\text{out}}-3I$ is nilpotent of index 2 on a 35-dimensional complement of $\operatorname{span}(C)$ and $\operatorname{rank}_{\mathbb{F}_7}(S-3I)\le 22$; this is consistent with the universal bound $\operatorname{rank}_{\mathbb{F}_7}(M)\ge 27$ established in [12] for $M=B^{\mathsf{T}}(3I-A)B+J\equiv -B^{\mathsf{T}}(A-3I-4J)B \pmod 7$ (B the vertex–triangle incidence). We found no contradiction.

4 Audit of the order-seven census (arXiv:2608.19410v1)

The 208 frequency formulas were transcribed from the TEX source. As printed, $\sum_i z_i-\binom{n}{7}=-\frac{1}{168}nk(k-2)(k-4)(19k^4-456k^3+4484k^2-21752k+42008)$ on the locus $n=k(k-2)/2+k+1$ ($-94\,034\,160$ at $(99,14)$). Matching the values at $n_3=705$ against the rank-207 affine family of [12] (wave 23) identifies exactly four defective formulas and their corrections: z_{21}: drop the factor $(k-3)$; z_{80},z_{84}: $(2k-7)\to(2k-17)$; z_{176}: $\frac{1}{84}\to\frac{1}{8}$. With these corrections $\sum_i z_i=\binom{n}{7}$ identically in k and all 208 formulas agree with the 208 classes of [12] in all three coefficients. The corrected census imposes only $0\le n_3\le 4158$, $2n_3\le z_{11}\le 4n_3$, $4\mid z_{11}$: no new information on n_3.

5 Computations

- Scripts: `orbit_sat.py` (CNF), `cpsat_orbit.py` (CP-SAT), both implementing (R),(D),(I),(P) for general $k=2m$, with a lift to the 99×99 matrix and an integer verification of $A^2+A-(k-2)I=2J$. Controls: $k=4$ gives SAT and the lifted matrix is $\operatorname{srg}(9,4,1,2)$ (Paley(9) has such an involution); $k=6$ gives UNSAT at once ($\operatorname{srg}(19,6,1,2)$ does not exist). These two controls are weak (the $k=4$ model has 3 variables). A strong control was added on 22 September: the Berlekamp–van Lint–Seidel graph $\operatorname{srg}(243,22,1,2)$ [11] with the involution $x\mapsto -x$ (one fixed point) is the $k=22$ instance of the same model, and its orbit table satisfies (R),(D),(I),(P) with zero violations. For $k=14$: 1722 primary variables, 666 183 CNF variables, 1 510 448 clauses.
- Sound case split on $D(U_0)$ for $U_0=(1,2,0)$: sibling $(1,2,1)$, one shared block $(1,3,0)$, disjoint $(3,4,0)$; for the disjoint case a further split on $u_0\sim\{a,b'\}$ (the block swap $1\leftrightarrow 2$ makes the two single-adjacency subcases equivalent).
- Results so far: signed-quotient relaxation (CP-SAT, 25 min): UNKNOWN. Joint model with the whole row of U_0 fixed: CaDiCaL undecided after ≥ 30 min CPU on four instances; CP-SAT 15 min: UNKNOWN (2 477 conflicts). Native CaDiCaL/Kissat batch (7 instances, 8 h each): all UNKNOWN (`results/BATCH_2026-09-21_overnight.md`). A timeout is not evidence.

6 What would settle the involution branch

An orderly generator on the 42 orbits using (i) interlacing of the partial symmetrized quotient against $\{14, 3^{27}, (-4)^{22}\}$, (ii) the matching D and the block constraints, (iii) Theorem 4 as a late filter, and (iv) the label group $2^7 \rtimes S_7$ (order 645 120) for isomorph rejection. Either outcome is certifiable: "no involution, hence $|\mathrm{Aut}(\Gamma)| \in \{1, 3\}$", or a Conway 99-graph.

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